5.1

$$f(x) = \log(x^4)\sin(x^3) = 4\log(x)\sin(x^3)$$

$$\frac{df}{dx} = \frac{d}{dx}\left(4\log(x)\right)\sin(x^3) + 4\log(x)\cdot\frac{d}{dx}\left(\sin(x^3)\right)$$

$$= \frac{4}{x}\sin(x^3) + 4\log(x)\cdot\cos(x^3)\cdot 3x^2$$

$$= \frac{4\sin(x^3)}{x} + 12x^2\log(x)\cos(x^3)$$

5.2

$$f(x) = \left(1+\exp(-x)\right)^{-1}$$

$$\frac{df}{dx} = \frac{d}{dx}\left(1+\exp(-x)\right)^{-1} =$$

$$= -\left(1+\exp(-x)\right)^{-2}\cdot\frac{d}{dx}\left(1+\exp(-x)\right) =$$

$$= -\left(1+\exp(-x)\right)^{-2}\cdot -\left(\exp(-x)\right) =$$

$$= \frac{\exp(-x)}{\left(1+\exp(-x)\right)^2}$$

5.3

$$\frac{d}{dx}\exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right) =$$

$$= \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)\cdot\frac{d}{dx}\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)$$

$$= \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)\cdot -\frac{1}{\cancel{2}\sigma^2}\cdot\cancel{2}(x-\mu) = -\frac{x-\mu}{\sigma^2}\exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)$$

(5.4)

$f^0(x) = \sin(x) + \cos(x)$ $\qquad f^0(0) = 1$

$f^1(x) = \cos(x) - \sin(x)$ $\qquad f^1(0) = 1$

$f^2(x) = -\sin(x) - \cos(x)$ $\qquad f^2(0) = -1$

$f^3(x) = -\cos(x) + \sin(x)$ $\qquad f^3(0) = -1$

$f^4(x) = \sin(x) + \cos(x)$ $\qquad f^4(0) = 1$

$f^5(x) = \cos(x) - \sin(x)$ $\qquad f^5(0) = 1$

$$T_5 = \sum_{k=0}^{5} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k = 1 + x + \left(-\frac{1}{2}x^2\right) + \left(-\frac{1}{6}x^3\right) + \frac{1}{24}x^4 + \frac{1}{120}x^5$$

$$= 1 + x - \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{24}x^4 + \frac{1}{120}x^5$$

$T_0 = 1$

$T_1 = 1 + x$

$T_2 = 1 + x - \frac{1}{2}x^2$

$T_3 = 1 + x - \frac{1}{2}x^2 - \frac{1}{6}x^3$

$T_4 = 1 + x - \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{24}x^4$

(5.5)

a) $\mathbb{R}^2 \to \mathbb{R}$ so dimension of $\frac{\partial f_1}{\partial x}$ is 1×2

$$\frac{\partial f_1}{\partial x} = \left[\frac{\partial f_1}{\partial x_1} \quad \frac{\partial f_1}{\partial x_2}\right] = \left[\cos(x_1)\cos(x_2) \quad -\sin(x_1)\sin(x_2)\right]$$

$$\frac{\partial f_1}{\partial x_1} = \frac{\partial}{\partial x_1}(\sin(x_1))\cos(x_2) + \underbrace{\frac{\partial}{\partial x_1}(\cos(x_2))}_{0}\sin(x_1) =$$

$$= \cos(x_1)\cos(x_2)$$

$$\frac{\partial f_1}{\partial x_2} = \underbrace{\frac{\partial}{\partial x_2}(\sin(x_1))}_{0}\cos x_2 + \frac{\partial}{\partial x_2}(\cos(x_2))\sin(x_1) = -\sin(x_1)\sin(x_2)$$

b)

$\mathbb{R}^n \to \mathbb{R}$ so $\frac{\partial f_2}{\partial x}$ is of dimension $1 \times \mathbb{R}^n$

$$x^T y = \sum_{i=1}^{n} x_i y_i \qquad \delta_{ij} \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$

$$\frac{\partial}{\partial x_j} \sum_{i=1}^{n} x_i y_i = \sum_{i=1}^{n} \frac{\partial}{\partial x_j}(x_i y_i) = \sum_{i=1}^{n} \delta_{ij}\, y_i = y_j$$

$$\frac{\partial f_2}{\partial x} = \left[\frac{\partial f_2}{\partial x_1} \cdots \frac{\partial f_2}{\partial x_n}\right] = \left[y_1 \cdots y_n\right] = \vec{y}^T$$

c)

$\mathbb{R}^n \to \mathbb{R}^{n \times n}$ so $\frac{\partial f_3}{\partial x}$ is $(n \times n) \times n$

$$x x^T = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \left[x_1 \cdots x_n\right] = \begin{bmatrix} x_1^2 & x_1 x_2 & \cdots & x_1 x_n \\ \vdots & x_2^2 & \ddots & \vdots \\ x_n x_1 & \cdots & & x_n^2 \end{bmatrix} = A \quad \text{where}$$

$A_{ij} = x_i x_j$

$i = 1 \ldots n$

$j = 1 \ldots n$

$$\frac{\partial A_{ij}}{\partial x_k} = \frac{\partial}{\partial x_k}(x_i x_j) =$$

$$= \frac{\partial}{\partial x_k}(x_i)\, x_j + \frac{\partial}{\partial x_k}(x_j)\, x_i =$$

$$= \delta_{ki}\, x_j + \delta_{kj}\, x_i$$

$$\frac{\partial A}{\partial x_1} \left[\quad \right] \frac{\partial A}{\partial x_2} \cdots \left[\quad \right] \frac{\partial A}{\partial x_n}$$

5.6

$$f(t) = \sin(\log(t^T t)) \quad , t \in \mathbb{R}^D \quad f(t) \in \mathbb{R}^1 \quad , \text{so } \frac{df}{dt} \in \mathbb{R}^1 \times \mathbb{R}^D$$

$$\frac{\partial}{\partial t_i} \sin\left(\log\left(\sum_{j=1}^{D} t_j^2\right)\right) = \cos\left(\log\left(\sum_{j=1}^{D} t_j^2\right)\right) \cdot \frac{1}{\sum_{j=1}^{D} t_j^2} \cdot \sum_{j=1}^{D} \frac{\partial}{\partial t_i} t_j^2$$

$$= \cos\left(\log\left(\sum_{j=1}^{D} t_j^2\right)\right) \cdot \frac{1}{\sum_{j=1}^{D} t_j^2} \cdot \sum_{j=1}^{D} 2 t_j \cdot \frac{\partial}{\partial t_i}(t_j) \qquad \delta_{ij} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$

$$= \cos\left(\log\left(\sum_{j=1}^{D} t_j^2\right)\right) \cdot \frac{1}{\sum_{j=1}^{D} t_j^2} \cdot \underbrace{\sum_{j=1}^{D} 2 t_j \delta_{ij}}_{\text{all other terms when } i \neq j \text{ will be } 0}$$

$$= \cos\left(\log\left(\sum_{j=1}^{D} t_j^2\right)\right) \cdot \frac{2 t_i}{\sum_{j=1}^{D} t_j^2}$$

$$= 2 \frac{t_i}{t^T t} \cos(\log(t^T t))$$

$$\frac{df}{dt} = \left[2 \frac{t_1}{t^T t} \cos(\log(t^T t)) \cdots 2 \frac{t_D}{t^T t} \cos(\log(t^T t)) \right] = \frac{2 t^T}{t^T t} \cos(\log(t^T t))$$

$$g(X) = AXB \qquad A \in \mathbb{R}^{D \times E}, X \in \mathbb{R}^{E \times F}, B \in \mathbb{R}^{F \times D}$$

$$g \in \mathbb{R}^1, \; X \in \mathbb{R}^{E \times F}, \text{ so } \frac{dg}{dX} \in 1 \times (\mathbb{R}^{E \times F})$$

$$Z = AX \quad \text{then} \quad z_{ik} = \sum_{j=1}^{E} a_{ij} \cdot x_{jk}$$

$$T = \begin{bmatrix} z_{11} & \cdots & z_{1k} \\ \vdots & & \\ z_{i1} & \cdots & z_{ik} \end{bmatrix} \begin{bmatrix} b_{11} & \cdots & b_{1l} \\ \vdots & & \\ b_{k1} & \cdots & b_{kl} \end{bmatrix}$$

$$t_{il} = \sum_{k=1}^{F} z_{ik} \cdot b_{kl} = \sum_{k=1}^{F} \left(\sum_{j=1}^{E} a_{ij} \cdot x_{jk} \right) \cdot b_{kl}$$

The trace is the sum of the digonal of T with elements t_{ii}

$$t_{ii} = \sum_{k=1}^{F} b_{ki} \cdot \sum_{j=1}^{E} a_{ij} \cdot x_{jk}$$

$$tr(T) = \sum_{i=1}^{D} t_{ii} = \sum_{i=1}^{D} \sum_{k=1}^{F} b_{ki} \cdot \sum_{j=1}^{E} a_{ij} \cdot x_{jk}$$

$$\frac{\partial tr(T)}{\partial x_{pq}} = \sum_{i=1}^{D} \sum_{k=1}^{F} b_{ki} \cdot \sum_{j=1}^{E} a_{ij} \cdot \frac{\partial}{\partial x_{pq}} x_{jk}$$

$$= \sum_{i=1}^{D} \sum_{k=1}^{F} b_{ki} \cdot \sum_{j=1}^{E} a_{ij} \cdot \delta_{pj} \cdot \delta_{qk}$$

So the only terms that are non-zero occur when

$p = j$ and $q = k$

where $p = 1, \ldots, E$

$q = 1, \ldots, F$

$$\frac{\partial tr(T)}{\partial x_{pq}} = \sum_{i=1}^{D} b_{qi} \cdot a_{ip}$$

$$= B_{q,:} \cdot A_{:,p}$$

denoting row q of B and column p of A

or

column q of B^T and row p of A^T

$$= A^T_{p,:} \cdot B^T_{:,q}$$

So $\dfrac{d(g(X))}{d(X)} = A^T B^T$

5.7

a) $f(z) = \log(1+z)$

$$\frac{df}{dx} = \frac{df}{dz} \cdot \frac{dz}{dx} \quad , \quad \frac{df}{dz} \in \mathbb{R}^1 \quad , \quad \frac{dz}{dx} \in \mathbb{R}^1 \times \mathbb{R}^D$$

$$\frac{df}{dz} = \frac{d}{dz} \log(1+z) = \frac{1}{1+z}$$

$$\frac{\partial z}{\partial x_i} = \frac{\partial}{x_i} \sum_{j=1}^{D} x_j^2 = \sum_{j=1}^{D} 2x_j \cdot \frac{\partial}{\partial x_i} x_j = \sum_{j=1}^{D} 2x_j \delta_{ij} = 2x_j$$

$$\frac{df}{dx} = \frac{1}{1+z} \cdot 2x^T = \frac{2x^T}{1+x^Tx}$$

b)

$f(z) = \sin(z) \qquad z = Ax+b$

$$\frac{df}{dz} \in \mathbb{R}^E \times \mathbb{R}^E \qquad \frac{dz}{dx} \in \mathbb{R}^E \times \mathbb{R}^D$$

$$\frac{df}{dx} \in \mathbb{R}^E \times \mathbb{R}^D$$

$$\frac{\partial f_i}{\partial z_j} = \delta_{ij} \cos(z_j) \longrightarrow \frac{df}{dz} = \begin{bmatrix} \cos(z_1) & 0 & \cdots & 0 \\ 0 & \cos(z_2) & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \cos(z_j) \end{bmatrix}$$

$$z_i = b_i + \sum_{j=1}^{D} a_{ij} x_j \qquad \frac{\partial z_i}{\partial x_k} = \frac{\partial}{\partial x_k}\left[b_i + \sum_{j=1}^{D} a_{ij} x_j\right]$$

$$\frac{\partial}{\partial x_k}\left[b_i + \sum_{j=1}^{D} a_{ij} x_j\right] = 0 + \sum_{j=1}^{D} a_{ij} \cdot \delta_{kj} = a_{ik}$$

so $\frac{dz}{dx} = A$

$$\underset{E \times E}{\begin{bmatrix} \cos(z_1) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \cos(z_i) \end{bmatrix}} \underset{E \times D}{\begin{bmatrix} a_{11} & \cdots & a_{1j} \\ \vdots & & \\ a_{i1} & \cdots & a_{ij} \end{bmatrix}} =$$

$$= \begin{bmatrix} \cos(z_1) a_{11} & \cos(z_1) a_{12} & \cdots & \cos(z_1) a_{1j} \\ \cos(z_2) a_{21} & \vdots & & \cos(z_2) a_{2j} \\ & \vdots & & \vdots \\ \cos(z_i) a_{i1} & \cdots & & \cos(z_i) a_{ij} \end{bmatrix}$$

5.8

a) $\frac{df}{dz} \in \mathbb{R}^{1 \times 1}$

$\frac{dz}{dy} \in \mathbb{R}^{1 \times D}$

$\frac{dy}{dx} \in \mathbb{R}^{D \times D}$

$$\frac{df}{dz} = \frac{d}{dz}\left(\exp\left(-\frac{1}{2} z\right)\right) =$$

$$= \exp\left(-\frac{1}{2} z\right) \cdot \frac{d}{dz}\left(-\frac{1}{2} z\right) =$$

$$= -\frac{1}{2} \exp\left(-\frac{1}{2} z\right)$$

$$z = \sum_{i=1}^{D} y_i \cdot \left(\sum_{j=1}^{D} s_{ij} \cdot y_j \right)$$

$$\frac{\partial z}{\partial y_k} = \frac{\partial}{\partial y_k} \left[\sum_{i=1}^{D} y_i \cdot \left(\sum_{j=1}^{D} s_{ij} \cdot y_j \right) \right]$$

$$= \sum_{i=1}^{D} \left[\left[\frac{\partial}{\partial y_k} (y_i) \cdot \sum_{j=1}^{D} s_{ij} \cdot y_j \right) + \left[y_i \cdot \frac{\partial}{\partial y_k} \left(\sum_{j=1}^{D} s_{ij} \cdot y_j \right) \right] \right]$$

$$= \sum_{i=1}^{D} \left[\left(\delta_{ik} \cdot \sum_{j=1}^{D} s_{ij} \cdot y_j \right) + \left(y_i \cdot \sum_{j=1}^{D} s_{ij} \cdot \frac{\partial}{\partial y_k} \cdot y_j \right) \right]$$

$$= \sum_{i=1}^{D} \left[\delta_{ik} \cdot \sum_{j=1}^{D} s_{ij} \cdot y_j + y_i \cdot \sum_{j=1}^{D} s_{ij} \cdot \delta_{kj} \right]$$

$$= \sum_{i=1}^{D} \left[\delta_{ik} \cdot \sum_{j=1}^{D} s_{ij} \cdot y_j + y_i \cdot s_{ik} \right] =$$

$$= \underbrace{\sum_{i=1}^{D} \left[\delta_{ik} \cdot \sum_{j=1}^{D} s_{ij} \cdot y_j \right]}_{\text{only if } i=k} + \sum_{i=1}^{D} y_i \cdot s_{ik} =$$

$$= \sum_{j=1}^{D} s_{kj} \cdot y_j + \sum_{i=1}^{D} y_i \cdot s_{ik} \qquad \text{swap } \sum_{i=1}^{D} \text{ for } \sum_{j=1}^{D}$$

$$= \sum_{j=1}^{D} s_{kj} \cdot y_j + \sum_{j=1}^{D} y_j \cdot s_{jk}$$

$$\frac{\partial z}{\partial y_k} = \sum_{j=1}^{D} s_{kj} \cdot y_j + y_j \cdot s_{jk} = \sum_{j=1}^{D} y_j \cdot (s_{kj} + s_{jk})$$

$$\frac{\partial z}{\partial y_k} = \sum_{j=1}^{D} y_j \cdot \left[(S^{-1})_{kj} + (S^{-1})_{jk} \right]$$

$$\frac{dz}{dy} = y^T \cdot \left[S^{-1} + (S^{-1})^T \right]$$

$$y_i = x_i - \mu_i \qquad \frac{\partial y_i}{\partial x_j} = \frac{\partial}{\partial x_j}(x_i - \mu_i) = \delta_{ij}$$

which in matrix form would be the identity matrix.

$$\frac{df}{dx} = \overbrace{-\frac{1}{2}\left(\exp\left(-\frac{1}{2}z\right)\right)}^{\mathbb{R}^{1\times 1}} \overbrace{y^T \left[S^{-1} + (S^{-1})^T \right]}^{\mathbb{R}^{1\times D}} \overset{\mathbb{R}^{D\times D}}{I}$$

$$\frac{df}{dx} \in \mathbb{R}^{1\times D}$$

☆ Could use eq. 5.107 for $\frac{dz}{dy}$ which simplifies things.

b)

$$xx^T + \sigma^2 I = \begin{bmatrix} x_1^2 + \sigma^2 & x_1 x_2 & \cdots & x_1 x_D \\ x_{21} & x_2^2 + \sigma^2 & & \vdots \\ \vdots & & \ddots & \vdots \\ x_{D1} & \cdots & \cdots & x_D^2 + \sigma^2 \end{bmatrix}$$

$f(x) \in \mathbb{R}^1$

$\frac{df}{dx} \in \mathbb{R}^{1\times D}$

$$\mathrm{tr}(xx^T + \sigma^2 I) = \sum_{i=1}^{D} x_i^2 + \sigma^2 = D\sigma^2 + \sum_{i=1}^{D} x_i^2$$

$$\frac{\partial f}{\partial x_j} = \frac{\partial}{\partial x_j}\left[D\sigma^2 + \sum_{i=1}^{D} x_i^2 \right] = 0 + \sum_{i=1}^{D} 2x_i \delta_{ij} = 2x_j$$

$$\frac{df}{dx} = \left[2x_1 \quad 2x_2 \quad \cdots \quad 2x_D \right] = 2x^T$$

c)

$$f = \tanh(z) \qquad \frac{df}{dz} \in \mathbb{R}^{M \times M}$$

$$f_i = \tanh(z_i)$$

$$\frac{\partial f_i}{\partial z_j} = \frac{\partial}{\partial z_j} \tanh(z_i) = \operatorname{sech}^2(z_i)\,\delta_{ij} \rightarrow \begin{bmatrix} \operatorname{sech}^2(z_1) & 0 & \cdots & 0 \\ 0 & \operatorname{sech}^2(z_2) & & \vdots \\ \vdots & & \ddots & \\ 0 & \cdots & & \operatorname{sech}^2(z_M) \end{bmatrix}$$

$$z = Ax + b$$

From 5.7 we know $\frac{dz}{dx} \in \mathbb{R}^{M \times N}$ $\quad \frac{dz}{dx} = A$

$$\frac{df}{dx} = \underbrace{\begin{bmatrix} \operatorname{sech}^2(z_1) & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & & \operatorname{sech}^2(z_M) \end{bmatrix}}_{\mathbb{R}^{M \times M}} A$$

5.9

$$\frac{d}{dv} g(z,v) = \frac{\partial g(z,v)}{\partial v} + \frac{\partial g(z,v)}{\partial z} \cdot \frac{\partial z}{\partial v}$$

$$\frac{\partial g(z,v)}{\partial v} = \frac{\partial}{\partial v}\left[\log p(x,z) - \log q(z,v)\right]$$

here we only look at the explicit dependence of v in $g(z,v)$.

$$= 0 - \frac{\partial}{\partial v} \log q(z,v) = -\frac{1}{q(z,v)} \cdot \frac{\partial}{\partial v} q(z,v)$$

$$\frac{\partial g(z,v)}{\partial z} = \frac{\partial}{\partial z}\left[\log p(x,z) - \log q(z,v)\right]$$

only look at explicit dependence of z in g.

$$= \frac{1}{p(x,z)} \cdot \frac{\partial p(x,z)}{\partial z} - \frac{1}{q(z,v)} \cdot \frac{\partial q(z,v)}{\partial z}$$

$$\frac{\partial z}{\partial v} = \frac{\partial t(e,v)}{\partial v}$$

$$\frac{d}{dv} g(z,v) = -\frac{1}{q(z,v)} \cdot \frac{\partial}{\partial v} q(z,v) + \left[\frac{1}{p(x,z)} \cdot \frac{\partial p(x,z)}{\partial z} - \frac{1}{q(z,v)} \cdot \frac{\partial q(z,v)}{\partial z}\right] \cdot \frac{\partial t(e,v)}{\partial v}$$

$$= \frac{-1}{q(z,v)} \cdot \frac{\partial}{\partial v} q(z,v) + \frac{1}{p(x,z)} \cdot \frac{\partial p(x,z)}{\partial z} \cdot \frac{\partial t(e,v)}{\partial v} - \frac{1}{q(z,v)} \cdot \frac{\partial q(z,v)}{\partial z} \cdot \frac{\partial t(e,v)}{\partial v}$$

$$= \frac{1}{p(x,z)} \cdot \frac{\partial p(x,z)}{\partial z} \cdot \frac{\partial t(e,v)}{\partial v} - \frac{1}{q(z,v)} \left[\frac{\partial}{\partial v} q(z,v) - \frac{\partial q(z,v)}{\partial z} \cdot \frac{\partial t(e,v)}{\partial v}\right]$$